Chapter 2

Background

[Your existing text: 2.1 Semantic Data Modeling ... 2.2 Knowledge Graphs, Definition 1, Example 1 stays unchanged.]

2.1 Semantic Data Modeling

(existing)

2.2 Knowledge Graphs

(existing text, Definition 1 and Example 1)

2.2.1 Querying Knowledge Graphs with SPARQL

Once knowledge is stored as RDF triples, a dedicated language is needed to retrieve it. The *SPARQL Protocol and RDF Query Language* (SPARQL) is the W3C standard for querying RDF data [1]. A SPARQL query describes a small graph pattern, that is, a set of triples in which some positions are replaced by variables (written with a leading `?`). The query engine then searches the Knowledge Graph for all sub-graphs that match this pattern and returns the values bound to the variables. In addition to simple look-ups, SPARQL supports filtering (`FILTER`), optional patterns (`OPTIONAL`), aggregation (`COUNT`, `GROUP BY`) and federated queries over several endpoints (`SERVICE`). SPARQL is therefore the natural interface between a Knowledge Graph and the applications built on top of it, including the data preparation steps for language models described later in this chapter.

Definition 2: Basic Graph Pattern

Let V be an infinite set of variables disjoint from Con. A *triple pattern* is a tuple $t \in (Con \cup V) \times (Con \cup V) \times (Con \cup V)$, and a *basic graph pattern* P is a finite set of triple patterns. A *solution* of P over a Knowledge Graph $KG = (N, E, L)$ is a mapping $\mu : V \to Con$ such that replacing every variable v in P by $\mu(v)$ yields a set of triples that is contained in E, i.e. $\mu(P) \subseteq E$ [1, 2].

Example 2: SPARQL over the questionnaire Knowledge Graph

In the DigiStrucMed questionnaire Knowledge Graph, every patient answer is linked to the question it answers and to an answer category (e.g. *very_positive*, *neutral*, *negative*). The following query counts, for every question of the category *Quality*, how many participants selected each answer category.

```
PREFIX ds: <http://digistrucmed.org/questionnaire#>
SELECT ?questionDesc ?category (COUNT(DISTINCT ?resp) AS ?numPeople)
WHERE {
  ?question a ds:Question ;
            ds:hasQuestionCategory ds:Quality ;
            ds:QuestionDescription ?questionDesc .
  ?resp     a ds:Answered ;
            ds:hasAnswer ?answer .
  ?answer   ds:isAnswerTo ?question ;
            ds:hasAnswerCategory ?category .
}
GROUP BY ?questionDesc ?category
ORDER BY ?questionDesc ?category
```

The `WHERE` clause is a basic graph pattern with seven triple patterns; each result row is one solution μ, aggregated by `GROUP BY`.

2.2.2 Open-World Assumption and Knowledge Graph Incompleteness

Knowledge Graphs are almost never complete: guidelines are updated, sources disagree, and automatic extraction misses facts. How a system interprets a *missing* triple is therefore an important design decision. Relational databases usually follow the *Closed-World Assumption* (CWA), under which every fact that is not stored is considered false. Semantic Web standards such as RDF and OWL instead follow the *Open-World Assumption* (OWA) [2].

Definition 3: Open-World Assumption

Under the Open-World Assumption, a triple $\langle s, p, o\rangle \notin E$ is *unknown* rather than false: the Knowledge Graph $KG = (N, E, L)$ is regarded as an incomplete description of the domain, and the absence of a triple neither confirms nor refutes the corresponding statement [2].

Example 3: Missing is not false

Assume a guideline Knowledge Graph contains the triple `(sacubitril/valsartan, recommendedFor, HFrEF)`, but no triple that connects sacubitril/valsartan to arterial hypertension. Under the CWA, a system would conclude that the drug is *not* related to hypertension. Under the OWA, the system can only conclude that the graph *does not state* such a relation. For a patient-facing assistant this difference matters: an answer such as "this drug has nothing to do with blood pressure" would be an unjustified claim derived from incomplete data.

The OWA has two direct consequences for this thesis. First, it motivates *reasoning* over Knowledge Graphs, i.e. deriving facts that are not stated explicitly but follow from other facts (Section 2.4). Second, it complicates the evaluation of such reasoning, because a predicted fact that is missing from the graph is not necessarily wrong. Rule-mining systems therefore introduce intermediate assumptions, such as the Partial Completeness Assumption discussed in Section 2.4.1.

2.2.3 Knowledge Graph Enrichment with External Resources

A Knowledge Graph that is built from a single type of source, for example the text of clinical guidelines, typically contains surface forms such as abbreviations (*MRA*, *SGLT2i*) or spelling variants (*beta-blocker*, *beta blockers*) without any further description. *Knowledge Graph enrichment* links such entities to external, curated resources and imports additional facts about them. In the biomedical domain, three resources are particularly relevant:

- **Wikidata**, a free, collaboratively edited, multilingual Knowledge Graph that provides structured statements, synonyms (aliases) and identifiers for millions of entities, including diseases and drugs, through a public SPARQL endpoint [3].
- **The Unified Medical Language System (UMLS)**, which integrates more than one hundred biomedical vocabularies (e.g. SNOMED CT, MeSH, ICD) under shared concept identifiers [4].
- **DrugBank**, a curated knowledge base that combines detailed chemical, pharmacological and drug-target information [5].

Example 4: Enriching a guideline abbreviation

The guideline Knowledge Graph of this thesis stores the abbreviation *MRA* as a surface form extracted from guideline text. Enrichment adds its expansion and links the drug class to a Wikidata entity:

```
ds:entity/MRA  a ds:Abbreviation, ds:SurfaceForm ;
    ds:hasExpansion ds:entity/Mineralocorticoid%20receptor%20antagonist .

ds:entity/beta%20blocker  a ds:WikidataEntity, ds:SurfaceForm, ds:Drug ;
    ds:subclass_of ds:entity/medication .
```

Because all pipelines mint IRIs in the same namespace, a guideline mention and a Wikidata concept with identical text become *one* node. This is what allows facts from different sources to be combined in a single query.

Enrichment increases the coverage and connectivity of the graph, which is important both for answering patient questions and for mining rules that generalise across documents. At the same time, imported facts inherit the quality and the licence of their source, and automatic entity linking can introduce errors. Enriched facts should therefore keep their provenance (e.g. the source document or the Wikidata identifier), so that they can be traced back and verified by medical experts.

2.3 Language Models

(existing)

2.3.1 Pre-Trained Language Models

(existing)

2.3.2 Large Language Models

(existing)

2.3.3 Fine-Tuning

(existing)

2.3.4 Parameter-Efficient Fine-Tuning

Full fine-tuning updates *all* parameters of a model. For current LLMs with billions of parameters, this requires a large amount of GPU memory and produces a complete copy of the model for every task. *Parameter-efficient fine-tuning* (PEFT) methods avoid this by freezing

the pre-trained weights and training only a small number of additional parameters. The most widely used PEFT method is *Low-Rank Adaptation* (LoRA) [6].

LoRA is based on the observation that the weight change needed to adapt a pre-trained model to a new task has a low "intrinsic rank". Instead of learning a full update matrix, LoRA learns two small matrices whose product approximates it (Figure 2.1).

Definition 4: Low-Rank Adaptation (LoRA)

Let $W_0 \in \mathbb{R}^{d\times k}$ be a frozen pre-trained weight matrix of a linear layer. LoRA represents the task-specific update as $\Delta W = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$ and rank $r \ll \min(d, k)$. For an input x, the adapted layer computes

$$h \;=\; W_0 x + \frac{\alpha}{r} BAx,$$

where α is a constant scaling factor. Only A and B are trained; B is initialised with zeros so that training starts from the unchanged pre-trained model [6].

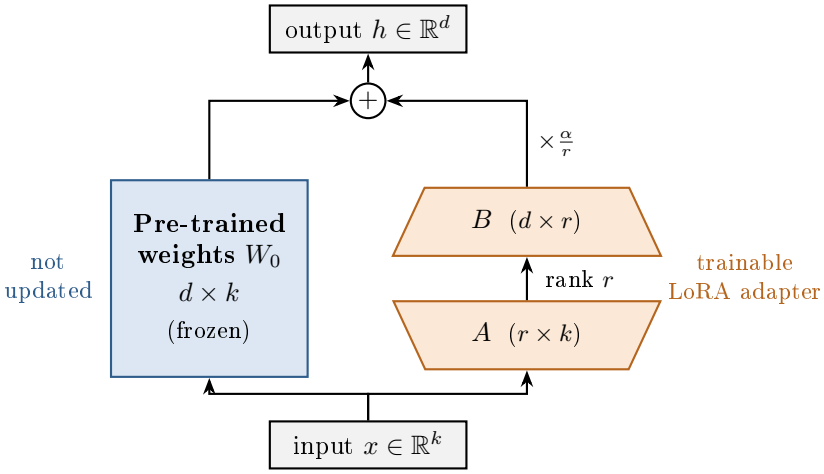


Figure 2.1: Low-Rank Adaptation (LoRA). The pre-trained weight matrix W_0 stays frozen; only the two small matrices A and B are trained, and their scaled product is added to the output of the original layer. Adapted from Hu et al. [6].

Example 5: How many parameters does LoRA train?

Consider a single attention projection of an 8-billion-parameter model with $d = k = 4096$. Full fine-tuning updates $d \cdot k = 4096 \cdot 4096 = 16{,}777{,}216$ parameters in this matrix. With LoRA of rank $r = 16$, only $r(d+k) = 16 \cdot (4096+4096) = 131{,}072$ parameters are trained, which is about 0.78 % of the original matrix. The resulting adapter can be stored as a small separate file and loaded on top of the unchanged base model.

QLoRA [7] further reduces memory requirements by storing the frozen base model in 4-bit precision while training the LoRA matrices in higher precision. This combination makes it possible to fine-tune models with billions of parameters, such as the Llama 3 family [8], on a single GPU. Because the base model remains unchanged, several adapters, for example one trained with and one trained without knowledge-graph rules, can be compared on top of exactly the same model, which is the experimental setting used later in this thesis.

2.3.5 Prompting, In-Context Learning and Chain-of-Thought

Besides changing the parameters of a model, its behaviour can be steered through its *input*. A *prompt* is the text that is given to an LLM to elicit a response; it typically combines an instruction, optional context (e.g. facts retrieved from a Knowledge Graph) and the actual question. Brown et al. [9] showed that sufficiently large models can solve new tasks from a few demonstrations placed in the prompt, without any parameter update. This ability is called *in-context learning*, and a prompt that contains demonstrations is called a *few-shot* prompt.

For tasks that require several reasoning steps, it helps to ask the model to write down its intermediate steps before answering.

Definition 5: Chain-of-Thought (CoT)

A *chain of thought* is a sequence of intermediate natural-language reasoning steps $s_1, \ldots, s_m$ that leads from an input question q to a final answer a. *Chain-of-thought prompting* provides demonstrations of the form $(q, s_1, \ldots, s_m, a)$ so that the model generates its own reasoning steps before producing the answer [10].

Chain-of-thought demonstrations can be used in two ways: at inference time inside the prompt, as originally proposed by Wei et al. [10], or as *training data* for supervised fine-tuning, so that the model learns to produce explicit reasoning by itself. The second option is the one followed in this thesis, where the reasoning steps are derived from rules over a Knowledge Graph (Section 2.4.2).

Example 6: Standard answer vs. chain-of-thought answer

Question: *"My ejection fraction is 25 %. Which type of heart failure do I have?"*

Standard answer: *"You have heart failure with reduced ejection fraction."*

Chain-of-thought answer:

1. The left-ventricular ejection fraction (LVEF) describes how much blood the left ventricle pumps out with each beat.
2. The ESC guideline classifies heart failure with an LVEF of $\leq 40\,\%$ as heart failure with reduced ejection fraction (HFrEF) [11].
3. $25\,\% \leq 40\,\%$.
4. *Therefore*, an LVEF of 25 % corresponds to HFrEF. The exact diagnosis is made by the treating cardiologist.

Both answers have the same conclusion, but only the second one makes the underlying medical knowledge and the reasoning step visible, so that a clinician can check each step.

2.3.6 Hallucination in Large Language Models

LLMs generate text by repeatedly predicting likely next tokens. Fluency is therefore not a guarantee of correctness: a model can produce statements that sound convincing but are not supported by any source. This phenomenon is known as *hallucination* [12].

Definition 6: Hallucination

A *hallucination* is generated content that is nonsensical or unfaithful to the provided source content or to established world knowledge. It is *intrinsic* if the output contradicts

the source, and *extrinsic* if the output cannot be verified from the source, i.e. it can neither be supported nor contradicted by it [12].

Example 7: A hallucination in patient education

A patient asks: *"My heart is functioning at 25 %. Am I eligible for a heart transplant?"* A model answers: *"Yes. Every patient with an ejection fraction below 30 % qualifies for a heart transplant."* The answer is fluent and contains a plausible-looking threshold, but no guideline states such a rule; eligibility depends on many additional clinical factors [11]. If the prompt contained a guideline excerpt describing the actual eligibility criteria, the answer contradicts its source and is an *intrinsic* hallucination; if no such source was given, the claim is unsupported and cannot be verified, which makes it an *extrinsic* hallucination.

In medicine, hallucinations are especially critical, because patients may act on incorrect information and often lack the background to detect errors [13]. Two complementary strategies are used to reduce them: (i) *grounding* the model in curated, explicit knowledge, such as a Knowledge Graph built from clinical guidelines (Section 2.5), and (ii) making the model's reasoning explicit, so that each step can be checked against that knowledge (Section 2.4). This thesis combines both strategies.

2.4 Neuro-Symbolic Artificial Intelligence

Artificial intelligence has developed along two traditions. *Symbolic* approaches represent knowledge explicitly, for example as ontologies, logical rules or Knowledge Graphs, and derive conclusions by transparent reasoning. They are interpretable and precise, but they are expensive to build and struggle with noisy, unstructured input such as free text. *Neural* (sub-symbolic) approaches, including LLMs, learn statistical patterns from large amounts of data. They are robust to noise and very good at producing language, but their knowledge is stored implicitly in millions of parameters and their conclusions are difficult to verify.

Neuro-symbolic AI (NeSy) aims to combine both traditions so that each compensates for the weaknesses of the other [14, 15]. Kautz [15] distinguishes several integration patterns, ranging from neural models that call a symbolic solver, to symbolic knowledge that is *compiled into* the training data or the loss function of a neural network. This thesis follows the latter pattern: symbolic rules learnt from a Knowledge Graph are translated into natural-language training examples, so that the neural model internalises the explicit knowledge during fine-tuning.

2.4.1 Rule Mining over Knowledge Graphs

A Knowledge Graph contains many regularities that are not stated as rules but can be learnt from the data, for example "if two patients answered the same question, then …". *Rule mining* discovers such regularities automatically and expresses them as logical rules.

Definition 7: Horn Rule

An *atom* is an expression $p(X, Y)$ where $p \in L$ is an edge label (predicate) and X, Y are variables or constants. A *Horn rule* has the form

$$\underbrace{B_1 \wedge B_2 \wedge \cdots \wedge B_n}_{\text{body}} \Rightarrow \underbrace{r(x, y)}_{\text{head}},$$

where the body is a conjunction of atoms and the head is a single atom. A rule is *closed* if every variable appears at least twice, which ensures that it makes concrete predictions. Replacing all variables by constants such that every body atom is a triple in E is called an *instantiation* of the rule; the instantiated head is a *prediction* [16, 17].

Not every rule that can be formed is useful, so candidate rules are ranked by quality measures. Under the Open-World Assumption, a prediction that is missing from the graph cannot simply be counted as an error (Section 2.2.2). The AMIE family of rule miners therefore introduced the *Partial Completeness Assumption* (PCA): if the graph knows *at least one* object y' for a subject x and relation r, it is assumed to know *all* of them; only in this case is a missing prediction $r(x, y)$ treated as a counter-example [16].

Definition 8: Rule Quality Measures

Let $\vec{B} \Rightarrow r(x, y)$ be a closed Horn rule and let $\vec{z}$ denote the variables of the body other than x and y.

$$supp(\vec{B} \Rightarrow r(x, y)) = \#\,(x, y) : \exists \vec{z} : \vec{B} \wedge r(x, y)$$

$$hc(\vec{B} \Rightarrow r(x, y)) = \frac{supp(\vec{B} \Rightarrow r(x, y))}{\#\,(x', y') : r(x', y')}$$

$$conf(\vec{B} \Rightarrow r(x, y)) = \frac{supp(\vec{B} \Rightarrow r(x, y))}{\#\,(x, y) : \exists \vec{z} : \vec{B}}$$

$$conf_{\,PCA}(\vec{B} \Rightarrow r(x, y)) = \frac{supp(\vec{B} \Rightarrow r(x, y))}{\#\,(x, y) : \exists \vec{z}, y' : \vec{B} \wedge r(x, y')}$$

The *support* counts correct predictions, the *head coverage* (hc) normalises the support by the size of the head relation, the *standard confidence* treats every missing prediction as false, and the *PCA confidence* only counts predictions as false if the graph knows another object for the same subject [16, 17].

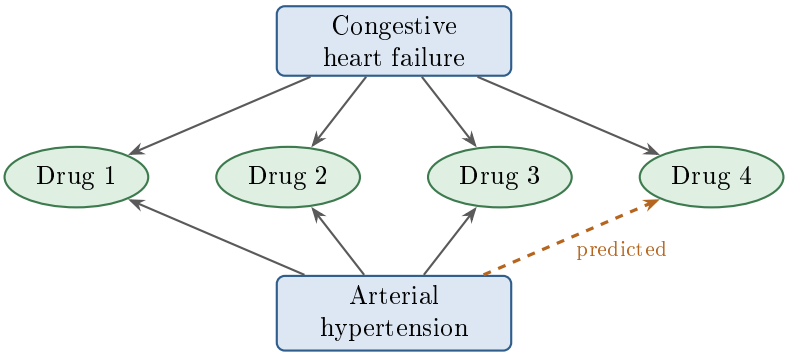
AMIE 3 [18] is the current version of this rule miner. It explores the space of closed Horn rules with a refinement strategy, prunes rules whose head coverage falls below a threshold, and computes all measures exactly while scaling to graphs with millions of facts. Figure 2.2 illustrates how a rule is supported by a Knowledge Graph and how it predicts a missing fact.

Example 8: A rule mined from the guideline Knowledge Graph

Running AMIE 3 on the guideline Knowledge Graph of this thesis (enriched with Wikidata) produces, among others, the rule

$$drug_or_therapy_used_for_treatment(congestive_heart_failure, b)$$
$$\Rightarrow drug_or_therapy_used_for_treatment(arterial_hypertension, b)$$

with $supp = 52$, a body size of 65, and therefore $conf = conf_{\,PCA} = 52/65 = 0.80$; its head coverage is 0.032. In words: *80 % of the drugs that the graph lists for congestive heart failure are also listed for arterial hypertension.* This reflects a real pharmacological overlap (e.g. diuretics, ACE inhibitors and beta-blockers are used for both conditions). However, a mined rule is a *statistical regularity of the graph*, not a clinical recommendation: the remaining 20 % of drugs show that the rule does not hold in general, and its predictions must not be presented to patients without verification.



Rule: $treatedBy(CHF, b) \Rightarrow treatedBy(AH, b)$
Solid edges: facts in the graph ($treatedBy$). Drugs 1–3 satisfy body *and* head (support = 3); Drug 4 satisfies only the body.
$conf = 3/4 = 0.75$. The dashed edge is the rule's prediction for the missing fact.

Figure 2.2: Rule mining over a toy Knowledge Graph. The rule is learnt from drugs that treat both diseases and is then used to predict a missing link (dashed). *treatedBy* abbreviates the Wikidata-derived relation *drug_or_therapy_used_for_treatment*.

2.4.2 From Symbolic Rules to Chain-of-Thought Training Data

Mined rules are formulated in the vocabulary of the Knowledge Graph and cannot be used directly to train a language model. They are therefore *grounded* in concrete entities and *verbalised* as natural language.

Definition 9: Rule Grounding

Let $\rho : \vec{B} \Rightarrow r(x, y)$ be a Horn rule. A *grounding* of ρ is a substitution θ that maps every variable of ρ to a constant in N such that every grounded body atom $\theta(B_i)$ is a triple in E. The grounded head $\theta(r(x, y))$ is the conclusion supported by the grounded body, which serves as its evidence or *reasoning path* [17, 19].

A grounding provides exactly the ingredients of a chain-of-thought demonstration (Definition of Chain-of-Thought in Section 2.3.5): the grounded body atoms are the intermediate reasoning steps, the rule is the justification that connects them, and the grounded head is the answer. Rules with low PCA confidence can be excluded or presented as uncertain, so that the model only learns reasoning patterns that are well supported by the data.

Example 9: From a rule to a chain-of-thought training instance (simplified)
Rule (guideline KG, $conf_{PCA} = 1.0$):

$$gradingSystemName(a, \text{“ESC class of recommendation”}) \Rightarrow gradeDimension(a, CLASS_OF_RECOMMENDATION)$$

Grounding: $a \mapsto g_{17}$, a grade record attached to one guideline recommendation.
Generated training instance:
Question: Which grade dimension does grade record g_{17} describe?
Reasoning: (1) The knowledge graph states that g_{17} uses the grading system “ESC class of recommendation”. (2) Records graded with this system describe the dimension *class of recommendation* (rule confidence 1.0). (3) Therefore, g_{17} describes a class of recom-

mendation.
Answer: CLASS_OF_RECOMMENDATION

2.5 Knowledge-Enhanced Large Language Models

Knowledge Graphs and LLMs have complementary strengths: Knowledge Graphs are explicit, verifiable and easy to update, while LLMs are flexible and produce fluent language. Pan et al. [20] distinguish three ways of combining them: (i) *KG-enhanced LLMs*, which use a Knowledge Graph during pre-training, fine-tuning or inference to improve the factual quality of the model; (ii) *LLM-augmented KGs*, which use LLMs to build, complete or query Knowledge Graphs; and (iii) *synergised* approaches, in which both components improve each other. This thesis belongs mainly to the first category: rules from the Knowledge Graph shape the fine-tuning data, and facts from the Knowledge Graph are given to the model as context at inference time.

2.5.1 Retrieval-Augmented Generation

The most common way of providing knowledge at inference time is *Retrieval-Augmented Generation* (RAG) [21].

Definition 10: Retrieval-Augmented Generation

Given an input x and a knowledge source $\mathcal{D}$, a retriever selects a set of relevant items $Z = \{z_1, \ldots, z_k\} \subseteq \mathcal{D}$, and a generator produces the output conditioned on both:

$$p(y \mid x) \approx \sum_{z \in Z} p_\eta(z \mid x)\, p_\theta(y \mid x, z),$$

where p_η is the retrieval model and p_θ the language model [21]. In *KG-RAG*, $\mathcal{D}$ is a Knowledge Graph and the retrieved items are triples, paths or sub-graphs that are serialised as text before being added to the prompt.

Because the knowledge is kept outside of the model, RAG allows updates without retraining, for example when a new version of a guideline is published, and the retrieved items can be shown to the user as evidence. However, KG-RAG implicitly assumes that the answer can be found in the graph. As discussed in Section 2.2.2, this is rarely true. Zhou et al. [19] therefore proposed BRINK, a benchmark in which the triple that directly answers a question is *removed*, while an alternative reasoning path remains. A system can only answer correctly if it *reasons* over the remaining facts instead of simply looking up the answer (Figure 2.3).

2.5.2 Evaluating Answers against Answer Sets

Many questions over a Knowledge Graph have more than one correct answer (e.g. "Which drugs are recommended for HFrEF?"). Model outputs are therefore compared with a *set* of gold answers rather than with a single string.

Definition 11: Answer-Set Metrics

For a question q, let A_q be the set of gold answers and P_q the set of answers extracted from the model output after normalisation. Then

$$\mathit{Hits@Any}(q) = \mathbb{1}\left[P_q \cap A_q \neq \emptyset\right], \quad \mathit{Prec}(q) = \frac{|P_q \cap A_q|}{|P_q|}, \quad \mathit{Rec}(q) = \frac{|P_q \cap A_q|}{|A_q|},$$

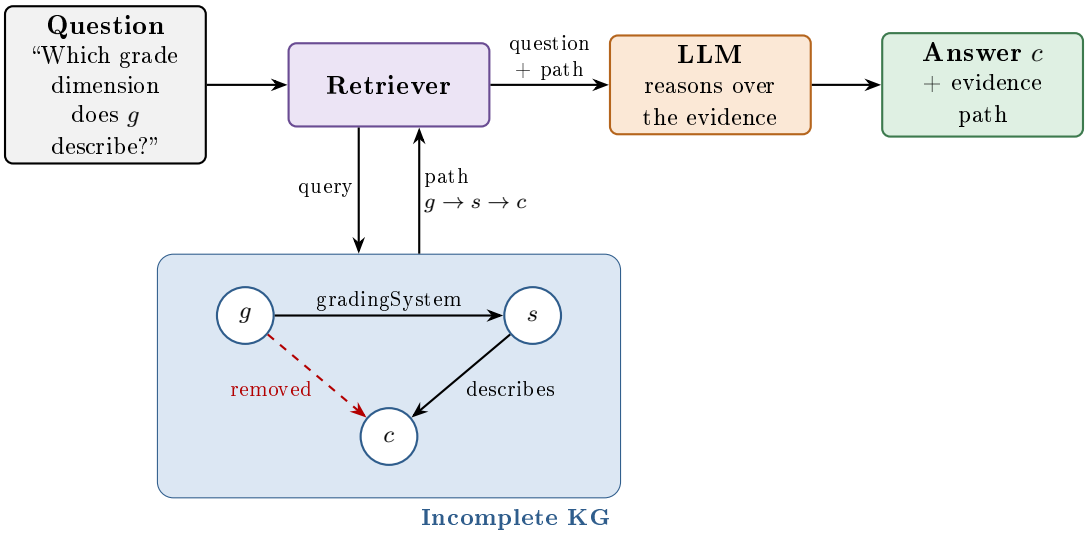


Figure 2.3: KG-based Retrieval-Augmented Generation under incomplete knowledge. The directly supporting triple ($g \to c$, red) is missing, so the LLM must infer the answer from the alternative path $g \to s \to c$. This is the evaluation setting of the BRINK benchmark [19].

$$F_1(q) = \frac{2 \cdot Prec(q) \cdot Rec(q)}{Prec(q) + Rec(q)}.$$

Hits@Hard additionally checks whether P_q contains the specific *hard* answer whose supporting triple was removed. All metrics are averaged over the questions of a test set [19].

Example 10: Computing the answer-set metrics

Let $A_q = \{\text{ACE inhibitor}, \text{ARNI}, \text{beta-blocker}, \text{MRA}\}$ and suppose the model answers "beta-blocker, MRA, digoxin", i.e. $P_q = \{\text{beta-blocker}, \text{MRA}, \text{digoxin}\}$. Then $|P_q \cap A_q| = 2$, so $Hits@Any = 1$, $Prec = 2/3 \approx 0.67$, $Rec = 2/4 = 0.50$ and $F_1 = (2 \cdot 0.67 \cdot 0.50)/(0.67 + 0.50) \approx 0.57$. The model found some correct answers (high Hits@Any), but it missed half of them and added one that is not in the gold set.

2.6 Automatic Prompt Optimization

The output of an LLM depends strongly on the wording of its prompt (Section 2.3.5). Writing good prompts by hand is a trial-and-error process that does not scale and is hard to reproduce. *Automatic prompt optimization* treats the prompt itself as the object that is optimised, while the model stays fixed.

Definition 12: Prompt Optimization Problem

Let $\mathcal{M}$ be a fixed language model, $\mathcal{P}$ the space of possible prompts, D_{val} a set of examples (x, y^*) and $f(\cdot) \in [0, 1]$ a task-specific scoring function. Prompt optimization searches for

$$p^* = \arg\max_{p \in \mathcal{P}} \frac{1}{|D_{val}|} \sum_{(x, y^*) \in D_{val}} f\big(\mathcal{M}(p, x),\, y^*\big).$$

Because $\mathcal{P}$ is discrete and $\mathcal{M}$ is used as a black box, gradient-based optimisation is not possible, and search heuristics are used instead [22, 23, 24].

Several strategies for this search have been proposed:

- **Automatic Prompt Optimization (APO/ProTeGi)** [22] evaluates a prompt on a mini-batch, asks an LLM to write a natural-language critique of the errors (a "textual gradient"), and then asks it to edit the prompt in the opposite direction of this critique.
- **Optimization by PROmpting (OPRO)** [23] shows an optimiser LLM a list of previous prompts together with their scores and asks it to propose a new prompt that is expected to score higher.
- **Genetic algorithms** [25, 26] maintain a *population* of candidate solutions and improve it over *generations* by *mutation* (random changes), *crossover* (combining two parents) and *selection* (keeping the fittest candidates).

GAAPO (Genetic Algorithmic Applied to Prompt Optimization) [24] combines these ideas. It is a genetic algorithm whose new candidates are produced by several different generators: LLM-based mutation with different strategies (e.g. adding an expert persona, decomposing the task, making the prompt more concise), crossover of two parent prompts, few-shot augmentation with training examples, APO and OPRO. All candidates are scored on the same validation set, and the best ones are kept as parents for the next generation (Figure 2.4). Since every candidate records which generator produced it, GAAPO also reveals which optimisation strategy works best for a given task.

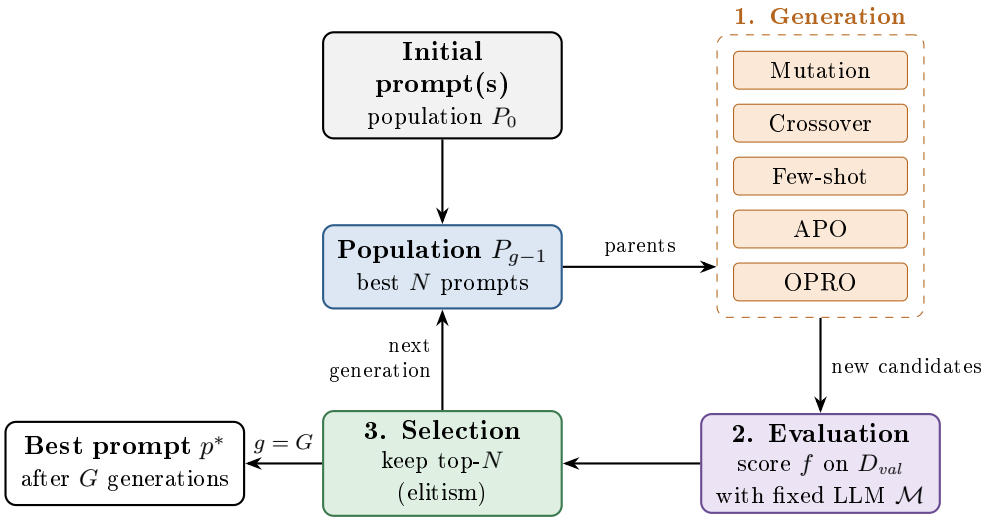


Figure 2.4: The GAAPO optimisation loop. In each generation, five generators create new prompt candidates from the current parents; all candidates are scored with the same fixed model and validation set, and the best N are kept. Based on Sécheresse et al. [24].

Example 11: Crossover and mutation of two prompts

Parent A: *"You are a cardiology nurse. Write one short video scene for patients. Use simple words."*

Parent B: *"Create a storyboard scene in JSON. Explain abbreviations at first mention."*

Crossover (first half of A + second half of B): *"You are a cardiology nurse. Write one short video scene. Explain abbreviations at first mention."*

Mutation (strategy "constraint addition"): *"... Keep the narration under 35 words."*

Both children are then scored, and they replace weaker members of the population only

if they achieve a higher score.

2.7 Cardiovascular Disease and Patient Education

This section introduces the medical application domain of the thesis. It is written for readers with a computer-science background and only covers the concepts needed to understand the data and the evaluation.

2.7.1 Cardiovascular Disease and Heart Failure

Cardiovascular diseases (CVDs) are a group of disorders of the heart and blood vessels, including coronary heart disease, cerebrovascular disease and heart failure. According to the World Health Organization, an estimated 19.8 million people died from CVDs in 2022, which corresponds to approximately 32 % of all global deaths [27]. Many of these deaths are related to modifiable risk factors such as high blood pressure, tobacco use, unhealthy diet and physical inactivity, which makes patient education an important part of prevention and long-term care.

Heart failure (HF) is not a single disease but a clinical syndrome with typical symptoms such as breathlessness, ankle swelling and fatigue. It is caused by a structural and/or functional abnormality of the heart, which leads to increased pressures inside the heart and/or an insufficient output of blood [11]. An important measurement is the *left-ventricular ejection fraction* (LVEF), the percentage of blood that the left ventricle ejects with each contraction. The European Society of Cardiology (ESC) uses the LVEF to distinguish three types of heart failure (Table 2.1).

Abbreviation	**Type of heart failure**	**LVEF**
HFrEF	HF with reduced ejection fraction	$\leq 40\,\%$
HFmrEF	HF with mildly reduced ejection fraction	41–49 %
HFpEF	HF with preserved ejection fraction	$\geq 50\,\%$

Table 2.1: Types of heart failure according to the 2021 ESC guidelines [11]. Additional criteria (symptoms, signs and further findings) are required for a diagnosis.

Patients with heart failure often have several co-morbidities, such as hypertension, diabetes or atrial fibrillation, and take several drugs at the same time. Their questions (“What does an ejection fraction of 25 % mean for me?”) therefore require answers that are correct, personalised and understandable at the same time, which is the motivating scenario of this thesis (see the motivating example in Chapter 1).

2.7.2 Clinical Practice Guidelines and Recommendation Grading

Clinical practice guidelines are documents in which expert panels summarise the available evidence and formulate recommendations for diagnosis and treatment. For heart failure in Europe, the main reference is the 2021 ESC guideline [11] together with its 2023 focused update [28]. Each recommendation is graded along two dimensions (Table 2.2): the *class of recommendation* expresses how strongly an action is recommended, and the *level of evidence* expresses how strong the underlying scientific evidence is.

Example 12: A graded recommendation as a Knowledge Graph

Dimension	Grade	Meaning / wording
Class of recommendation	Class I	Is recommended or indicated
	Class IIa	Should be considered
	Class IIb	May be considered
	Class III	Is not recommended
Level of evidence	Level A	Data from multiple randomised clinical trials or meta-analyses
	Level B	Data from a single randomised trial or large non-randomised studies
	Level C	Consensus of experts and/or small studies, retrospective studies, registries

Table 2.2: Grading of recommendations in ESC guidelines [11].

The guideline sentence *"An MRA is recommended for patients with HFrEF to reduce the risk of HF hospitalisation and death (Class I, Level A)"* (paraphrased from [11]) can be represented as a small graph: a node for the recommendation with edges to the drug class (*MRA*), the patient group (*HFrEF*), the outcome (*HF hospitalisation and death*), the source document, and a grade record that has a *classOfRecommendation* (*I*) and a *levelOfEvidence* (*A*). Keeping the grade and the source as explicit nodes allows a generated answer to state not only *what* is recommended, but also *how strongly* and *according to which guideline*.

2.7.3 Health Literacy and Readability

Medical correctness alone is not enough; patients must also be able to understand and use the information.

Definition 13: Health Literacy

Health literacy "entails people's knowledge, motivation and competences to access, understand, appraise, and apply health information in order to make judgments and take decisions in everyday life concerning healthcare, disease prevention and health promotion to maintain or improve quality of life during the life course" [29].

A simple, widely used proxy for how understandable a text is, is a *readability formula*. The best known is the Flesch Reading Ease score [30]; Amstad [31] adapted it to German, which is the language of the DigiLearnHF platform.

Definition 14: Flesch Reading Ease

Let ASL be the average sentence length (words per sentence) and ASW the average number of syllables per word. Then

$$FRE_{en} = 206.835 - 1.015 \cdot ASL - 84.6 \cdot ASW \qquad \text{(English, [30])}$$

$$FRE_{de} = 180 - ASL - 58.5 \cdot ASW \qquad \text{(German, [31])}$$

Higher values indicate easier texts; scores of 60–70 correspond to plain language and scores above 80 to easy texts [30].

Example 13: Same content, different readability

Technical: *"Pharmacological therapy with mineralocorticoid receptor antagonists reduces hospitalisation in patients with reduced ejection fraction."* (14 words, 44 syllables)
$FRE_{en} = 206.835 - 1.015 \cdot 14 - 84.6 \cdot \frac{44}{14} \approx -73$ (extremely difficult)

Plain: *"Some heart medicines lower the risk that you have to go to the hospital."* (14 words, 19 syllables)
$FRE_{en} = 206.835 - 1.015 \cdot 14 - 84.6 \cdot \frac{19}{14} \approx 78$ (fairly easy)

Both sentences have the same length; the difference comes entirely from long, technical words. Readability formulas do not measure correctness, so a simplified text must still be checked against the original guideline.

2.7.4 Educational Videos and Multimedia Learning

Short educational videos are a popular format for patient education. A systematic review of video-based interventions for patients with chronic illnesses found strong evidence that videos improve patients' knowledge of their condition, whereas effects on clinical outcomes were less consistent [32]. The design of such videos can be guided by the *cognitive theory of multimedia learning* [33], which assumes that people process information through two channels (visual and auditory), that each channel has a limited capacity, and that learning requires active processing. From this theory, Mayer derives design principles such as:

- **Multimedia principle:** people learn better from words and pictures than from words alone.
- **Coherence principle:** extraneous material (decorative pictures, unnecessary details) should be excluded.
- **Signalling principle:** key information should be highlighted, e.g. with short on-screen text.
- **Segmenting principle:** complex content should be presented in short, learner-paced segments (scenes).
- **Modality principle:** graphics are better explained with spoken narration than with long printed text.

Before a video is produced, its content is planned in a *storyboard*.

Definition 15: Storyboard

A *storyboard* is an ordered sequence of scenes $S = (s_1, \ldots, s_n)$ that specifies an educational video before production. Each scene $s_i = (narration_i, visual_i, text_i)$ consists of the spoken narration, a description of the visual content, and short on-screen text that signals the key message.

Example 14: A one-scene storyboard

The following storyboard, taken from the dataset used in this thesis, answers the request *"Create a short patient-education scene that explains the NYHA classification of heart failure and how it relates to symptom severity."*

```
{ "title": "Understanding NYHA Heart Failure Classes",
  "scenes": [ {
    "scene_number": 1,
```

```
"narration": "Heart failure is grouped into NYHA Classes I to IV,
              reflecting how symptoms limit daily activities and
              guide treatment decisions.",
"visual_description": "Simple chart showing four icons representing
                       increasing activity limitation.",
"on_screen_text": "NYHA I-IV: Symptom Levels" } ] }
```

The narration carries the explanation (modality principle), the visual illustrates it (multimedia principle), and the on-screen text highlights the key term (signalling principle).

A structured storyboard format has two advantages for LLM-based generation: it can be validated automatically (e.g. whether all fields are present and the narration is short enough), and its medical statements can be checked scene by scene against the Knowledge Graph before the video is produced.

2.8 Summary

Figure 2.5 summarises how the concepts introduced in this chapter relate to each other. Semantic data modelling and RML mappings turn questionnaires and clinical guidelines into Knowledge Graphs (Sections 2.2.1–2.2.3). Rule mining extracts symbolic regularities from these graphs, which are grounded and verbalised as chain-of-thought training data for parameter-efficient fine-tuning (Sections 2.3.4, 2.4). At inference time, the fine-tuned model receives Knowledge Graph context (Section 2.5) and a prompt that is optimised automatically (Section 2.6) to produce patient-oriented storyboards for educational videos (Section 2.7).

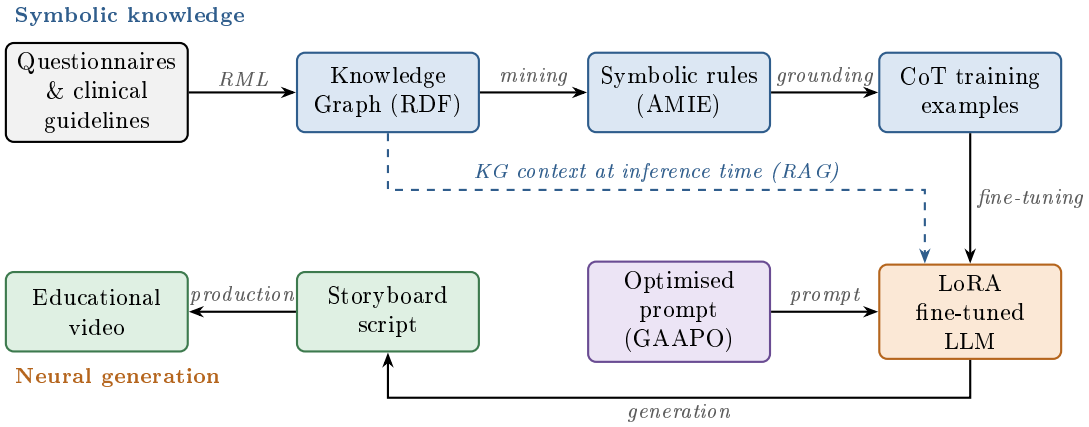


Figure 2.5: Overview of how the background concepts of this chapter are connected in the neuro-symbolic pipeline of this thesis.

Bibliography

[1] S. Harris and A. Seaborne, "SPARQL 1.1 query language." W3C Recommendation, Mar. 2013.

[2] A. Hogan, E. Blomqvist, M. Cochez, C. d'Amato, G. de Melo, C. Gutierrez, S. Kirrane, J. E. Labra Gayo, R. Navigli, S. Neumaier, A.-C. Ngonga Ngomo, A. Polleres, S. M. Rashid, A. Rula, L. Schmelzeisen, J. Sequeda, S. Staab, and A. Zimmermann, "Knowledge graphs," *ACM Computing Surveys*, vol. 54, no. 4, pp. 71:1–71:37, 2021.

[3] D. Vrandečić and M. Krötzsch, "Wikidata: A free collaborative knowledgebase," *Communications of the ACM*, vol. 57, no. 10, pp. 78–85, 2014.

[4] O. Bodenreider, "The unified medical language system (UMLS): Integrating biomedical terminology," *Nucleic Acids Research*, vol. 32, no. suppl_1, pp. D267–D270, 2004.

[5] D. S. Wishart, Y. D. Feunang, A. C. Guo, E. J. Lo, A. Marcu, J. R. Grant, *et al.*, "DrugBank 5.0: A major update to the DrugBank database for 2018," *Nucleic Acids Research*, vol. 46, no. D1, pp. D1074–D1082, 2018.

[6] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022.

[7] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[8] A. Grattafiori *et al.*, "The Llama 3 herd of models," 2024.

[9] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1877–1901, 2020.

[10] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 24824–24837, 2022.

[11] T. A. McDonagh, M. Metra, M. Adamo, R. S. Gardner, A. Baumbach, M. Böhm, *et al.*, "2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure," *European Heart Journal*, vol. 42, no. 36, pp. 3599–3726, 2021.

[12] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 248:1–248:38, 2023.

[13] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," *Nature Medicine*, vol. 29, no. 8, pp. 1930–1940, 2023.

[14] A. d'Avila Garcez and L. C. Lamb, "Neurosymbolic AI: The 3rd wave," *Artificial Intelligence Review*, vol. 56, no. 11, pp. 12387–12406, 2023.

[15] H. Kautz, "The third AI summer: AAAI Robert S. Engelmore memorial lecture," *AI Magazine*, vol. 43, no. 1, pp. 105–125, 2022.

[16] L. A. Galárraga, C. Teflioudi, K. Hose, and F. M. Suchanek, "AMIE: Association rule mining under incomplete evidence in ontological knowledge bases," in *Proceedings of the 22nd International Conference on World Wide Web (WWW)*, pp. 413–422, 2013.

[17] L. Galárraga, C. Teflioudi, K. Hose, and F. M. Suchanek, "Fast rule mining in ontological knowledge bases with AMIE+," *The VLDB Journal*, vol. 24, no. 6, pp. 707–730, 2015.

[18] J. Lajus, L. Galárraga, and F. Suchanek, "Fast and exact rule mining with AMIE 3," in *The Semantic Web – 17th International Conference, ESWC 2020*, vol. 12123 of *Lecture Notes in Computer Science*, pp. 36–52, Springer, 2020.

[19] D. Zhou, Y. Zhu, X. Wang, H. Zhou, Y. He, J. Chen, S. Staab, and E. Kharlamov, "What breaks knowledge graph based RAG? Benchmarking and empirical insights into reasoning under incomplete knowledge," in *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, (Rabat, Morocco), pp. 2522–2538, Association for Computational Linguistics, 2026.

[20] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 7, pp. 3580–3599, 2024.

[21] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[22] R. Pryzant, D. Iter, J. Li, Y. T. Lee, C. Zhu, and M. Zeng, "Automatic prompt optimization with "gradient descent" and beam search," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7957–7968, Association for Computational Linguistics, 2023.

[23] C. Yang, X. Wang, Y. Lu, H. Liu, Q. V. Le, D. Zhou, and X. Chen, "Large language models as optimizers," in *International Conference on Learning Representations (ICLR)*, 2024.

[24] X. Sécheresse, J.-Y. Guilbert-Ly, and A. Villedieu de Torcy, "GAAPO: Genetic algorithmic applied to prompt optimization," *Frontiers in Artificial Intelligence*, vol. 8, 2025.

[25] J. H. Holland, *Adaptation in Natural and Artificial Systems*. Ann Arbor, MI: University of Michigan Press, 1975.

[26] D. E. Goldberg, *Genetic Algorithms in Search, Optimization, and Machine Learning*. Reading, MA: Addison-Wesley, 1989.

[27] World Health Organization, "Cardiovascular diseases (CVDs) – fact sheet." https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds), July 2025. Accessed: 2026-10-16.

[28] T. A. McDonagh, M. Metra, M. Adamo, R. S. Gardner, A. Baumbach, M. Böhm, *et al.*, "2023 focused update of the 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure," *European Heart Journal*, vol. 44, no. 37, pp. 3627–3639, 2023.

[29] K. Sørensen, S. Van den Broucke, J. Fullam, G. Doyle, J. Pelikan, Z. Slonska, and H. Brand, "Health literacy and public health: A systematic review and integration of definitions and models," *BMC Public Health*, vol. 12, p. 80, 2012.

[30] R. Flesch, "A new readability yardstick," *Journal of Applied Psychology*, vol. 32, no. 3, pp. 221–233, 1948.

[31] T. Amstad, *Wie verständlich sind unsere Zeitungen?* PhD thesis, Universität Zürich, 1978.

[32] N. Deshpande, M. Wu, C. Kelly, N. Woodrick, D. A. Werner, A. Volerman, and V. G. Press, "Video-based educational interventions for patients with chronic illnesses: Systematic review," *Journal of Medical Internet Research*, vol. 25, p. e41092, 2023.

[33] R. E. Mayer, *Multimedia Learning*. Cambridge: Cambridge University Press, 3 ed., 2020.